Jing REN

UdeM – Pavillon André-Aisenstadt, Montreal, QC H3T 1N8, Canada
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Research Interests

computer graphics → geometry processing and digital fabrication → shape modeling and shape analysis, computational origami and kirigami, computational embroidery (smocking), non-rigid shape matching, shape deformation, urban reconstruction, etc.

Academic Positions

Assistant Professor *2026 - present*
Université de Montréal, Canada

Associate Academic Member *2026 - present*
MILA, Canada

Senior Researcher | *advised by Prof. Olga Sorkine-Hornung* *2023 - 2026*
ETH Zurich, Switzerland

Postdoctoral Researcher | *advised by Prof. Olga Sorkine-Hornung* *2021 - 2023*
ETH Zurich, Switzerland

Honors & Awards

Junior Fellow | *Eurographics Association* *2026 - present*

Eurographics Young Researcher Award *2026*

Best Paper Award honorable mention | *SIGGRAPH* *2024*

Best Paper Award | *International Conference on 3D Vision* *2022*

Student Research Excellence Award | *KAUST* *2020*

Best Paper Award honorable mention | *Symposium on Geometry Processing* *2019*

Graduate with distinction | *Oxford University* *2015*

Graduate with honors in Program of Science & Engineering | *Zhejiang University* *2014*

Scholarship for Outstanding Merits | *Zhejiang University* *2011-2013*

Student Excellence Award | *Zhejiang University* *2011*

Education

Ph.D. in Computer Science | *advised by Prof. Peter Wonka and Prof. Maks Ovsjanikov* *2015 - 2021*
KAUST, Saudi Arabia

M.Sc in Financial Mathematics *2014 - 2015*
University of Oxford, United Kingdom

B.Sc. in Applied Mathematics *2010 - 2014*
Zhejiang University, China

Publications

24 **Uniformly Deployable Kirigami on Arbitrary Planar Graphs**
Aviv Segall, Jing Ren, Olga Sorkine-Hornung
ACM Transactions on Graphics (Proc. SIGGRAPH), 2026

23 **Gradient Descent in the ALPS: Abstracted Low-Poly Stylization and Fabrication**
Ruben Wiersma*, Alexandre Binninger*, Peizhuo Li, Tanguy Magne, Annika Öhri, Aviv Segall, Danielle Luterbacher, Marcel Padilla, Jing Ren, Olga Sorkine-Hornung
SIGGRAPH, 2026 ∗equal contribution

22 **Reconfigurable Hinged Kirigami Tessellations**
Aviv Segall*, Jing Ren*, Marcel Padilla, Olga Sorkine-Hornung
SIGGRAPH Asia, 2025 ∗equal contribution

21 **Geometry Distributions**
Biao Zhang, Jing Ren, Peter Wonka
ICCV, 2025

20 **Computational Modeling of Gothic Microarchitecture**
Aviv Segall, Jing Ren, Martin Schwarz, Olga Sorkine-Hornung
SIGGRAPH, 2025

19 **Chebyshev Parameterization for Woven Fabric Modeling**
Annika Öhri*, Aviv Segall*, Jing Ren, Olga Sorkine-Hornung
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2024 ∗equal contribution

18 **Fabric Tessellation: Realizing Freeform Surfaces by Smocking**
Aviv Segall, Jing Ren, Amir Vaxman, Olga Sorkine-Hornung
ACM Transactions on Graphics (Proc. SIGGRAPH), 2024

17 **Computational Smocking through Fabric-Thread Interaction**
Ningfeng Zhou, Jing Ren, Olga Sorkine-Hornung
Eurographics, 2024

16 **Digital 3D Smocking Design**
Jing Ren, Aviv Segall, Olga Sorkine-Hornung
ACM Transactions on Graphics, 2024 (presented in SIGGRAPH Asia 2023)

15 **Smooth Non-Rigid Shape Matching via Effective Dirichlet Energy Optimization**
Robin Magnet, Jing Ren, Olga Sorkine-Hornung, Maks Ovsjanikov
International Conference on 3D Vision (3DV), 2022

14 **Learning to Construct 3D Building Wireframes from 3D Line Clouds**
Yicheng Luo, Jing Ren, Xuefei Zhe, Di Kang, Yajing Xu, Peter Wonka, Linchao Bao
British Machine Vision Conference (BMVC), 2022

13 **Gaussian Blue Noise**
Abdalla G. M. Ahmed, Jing Ren, Peter Wonka
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2022

12 **REALY: Rethinking the Evaluation of 3D Face Reconstruction**
Zenghao Chai*, Haoxian Zhang*, Jing Ren, Di Kang, Zhengzhuo Xu, Xuefei Zhe, Chun Yuan, Linchao Bao
European Conference on Computer Vision (ECCV), 2022 ∗equal contribution

11 **Intuitive and Efficient Roof Modeling for Reconstruction and Synthesis**
Jing Ren, Biao Zhang, Bojian Wu, Jianqiang Huang, Lubin Fan, Maks Ovsjanikov, Peter Wonka
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2021

10 **Discrete Optimization for Shape Matching**
Jing Ren, Simone Melzi, Peter Wonka, Maks Ovsjanikov

Computer Graphics Forum (Proc. SGP), 2021

9 **Fast Sinkhorn Filters: Using Matrix Scaling for Non-Rigid Shape Correspondence with Functional Maps**
Gautam Pai, Jing Ren, Simone Melzi, Peter Wonka, Maks Ovsjanikov
Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021

8 **Geometric analysis of shape variability of lower jaws of prehistoric humans**
Jing Ren, Peter Wonka, Gowtham Harihara, Maks Ovsjanikov
L'Anthropologie, 2020

7 **MapTree: Recovering Multiple Solutions in the Space of Maps**
Jing Ren, Simone Melzi, Maks Ovsjanikov, Peter Wonka
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2020

6 **MGCN: Descriptor Learning using Multiscale GCNs**
Yiqun Wang, Jing Ren, Dong-Ming Yan, Jianwei Guo, Xiaopeng Zhang, Peter Wonka
ACM Transactions on Graphics (Proc. SIGGRAPH), 2020

5 **Consistent ZoomOut: Efficient Spectral Map Synchronization**
Ruqi Huang, Jing Ren, Peter Wonka, Maks Ovsjanikov
Computer Graphics Forum (Proc. SGP), 2020

4 **ZoomOut: Spectral Upsampling for Efficient Shape Correspondence**
Simone Melzi*, Jing Ren*, Emanuele Rodolà, Abhishek Sharma, Peter Wonka, Maks Ovsjanikov
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2019 ∗equal contribution

3 **Structured Regularization of Functional Map Computations**
Jing Ren, Mikhail Panine, Peter Wonka, Maks Ovsjanikov
Computer Graphics Forum (Proc. SGP), 2019

2 **Continuous and Orientation-preserving Correspondence via Functional Maps**
Jing Ren, Adrien Poulenard, Peter Wonka, Maks Ovsjanikov
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2018

1 **Joint Graph Layouts for Visualizing Collections of Segmented Meshes**
Jing Ren, Jens Schneider, Maks Ovsjanikov, Peter Wonka
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2017

Academic Service

Computers & Graphics (Elsevier) | *Associate Editor* — *2026 - present*

ACM SIGGRAPH | *Technical Papers Committee Member* — *2023, 2024, 2026*

ACM SIGGRAPH Asia | *Technical Papers Committee Member* — *2025, 2026*

EUROGRAHPICS | *Technical Papers Committee Member* — *2025 - 2026*

Eurographics Symposium on Geometry Processing | *Technical Papers Committee Member* — *2023 - 2026*

Pacific Graphics | *Technical Papers Committee Member* — *2022, 2024 - 2026*

Eurographics Symposium on Geometry Processing | *Graduate School Co-chair* — *2025*

CAD/Graphics | *Technical Papers Committee Member* — *2025*

ACM SIGGRAPH | *Reviewer* — *2022 - 2025*

ACM SIGGRAPH Asia | *Reviewer* — *2023 - 2025*

EUROGRAHPICS | *Reviewer* — *2023 - 2026*

Eurographics Symposium on Geometry Processing | *Reviewer* — *2023 - 2026*

ACM Computer-Human Interaction | *Reviewer* *2026*

ACM Transactions on Graphics | *Reviewer* *2021 - 2024*

IEEE Transactions on Visualization and Computer Graphics | *Reviewer* *2021, 2024, 2026*

European Conference on Computer Vision (ECCV) | *Outstanding Reviewer Award* *2024*

Computer Graphics Forum | *Reviewer* *2022, 2024, 2025*

IEEE Computer Graphics and Applications | *Reviewer* *2018, 2025*

ACM Symposium on Computational Fabrication | *Reviewer* *2024*

International Conference on Machine Learning (ICML) | *Reviewer* *2023*

Journal of Photogrammetry and Remote Sensing (ISPRS) | *Top 10% Reviewer Recognition* *2023*

Computers & Graphics | *Reviewer* *2023*

Conference on Neural Information Processing Systems (NeurIPS) | *Reviewer* *2022*

British Machine Vision Conference (BMVC) | *Reviewer* *2022*

Teaching

Linear Algebra @ ETH Zurich — *teaching assistant, Autumn 2024, 2025, 2026*

Shape Modeling and Geometry Processing @ ETH Zurich — *guest lecturer, Spring 2023, 2024, 2025*

Computational Design for Additive Manufacturing (CDfAM) @ ETH Zurich — *summer school guest lecturer, Jul 2024*

Linear Algebra @ ETH Zurich — *head teaching assistant, Autumn 2022*

Shape Modeling and Geometry Processing @ ETH Zurich — *teaching assistant, Spring 2022*

2× **Master Thesis,** 2× **Bachelor Thesis,** 4× **Semester Project** @ ETH Zurich — *Thesis (co-)Advisor, 2022-Now*

Other Research Experience

Tencent, AI Lab (Digital Human) *Shenzhen, China*
Researcher *Jul 2021 - Dec 2021*

Alibaba, DAMO Academy, AI center (City Brain) *Zhejiang, China*
Research Intern *Jul 2020 - Jun 2021*

École Polytechnique, Laboratoire d'Informatique (LIX) *Palaiseau, France*
Research Visitor; advised by Prof. Maks Ovsjanikov *Jun - Aug 2017, 2018, 2019*

Invited Talks & Outreach Events (Selection)

Geometry Workshop in Obergurgl *Obergurgl, Austria*
Topic: "Realizing freeform surfaces from flat sheets via sewing and cutting" *Aug 31, 2025*

GAMES Webinar *online*
Topic: "Digital Smocking Design", hosted by Prof. Haisen Zhao *Jan 06, 2025*

Seminar talk @ Universität Siegen *Siegen, Germany*
Topic: "Shape matching and map space exploration via functional maps", hosted by Prof. Michael Möller *Mar 26, 2024*

NACHTAKTIV "Science Catwalk" @ Museum Rietberg *Zurich, Switzerland*
Topic: "Digital 3D Smocking Design" [event] *Nov 2, 2023*

Seminar talk @ IST Austria *Klosterneuburg, Austria*
Topic: "Digital 3D Smocking Design", hosted by Prof. Chris Wojtan *Oct 24, 2023*

Seminar talk @ TU Wien *Vienna, Austria*
Topic: "Digital 3D Smocking Design", hosted by Prof. Michael Wimmer *Oct 20, 2023*

Seminar talk @ LORIA *Nancy, France*
Topic: "Digital 3D Smocking Design", hosted by Dr. Étienne Corman *Oct 4, 2023*

Headline speaker @ Toronto Geometry Colloquium *online*
Topic: "Non-rigid Shape Matching via Functional Maps" *Sep 17, 2021*

Seminar talk @ Central South University *online*
Topic: "Non-rigid Shape Matching via Functional Maps", hosted by Prof. Shengjun Liu *Jun 27, 2021*

Seminar talk @ TBSI, Tsinghua University *online*
Topic: "ZoomOut: Spectral Upsampling for Efficient Shape Correspondence", hosted by Prof. Ruqi Huang *May 31, 2021*

Guest lecture @ ShanghaiTech University *online*
Topic: "Discrete Laplace-Beltrami Operator", hosted by Prof. Chi-Han Peng *May 7, 2020*